A-A-2538
October 14, 1987

COMMERCIAL ITEM DESCRIPTION

ADJUSTING TOOL, VALVE TAPPET

The General Services Adminstration has authorized the use of this commercial item description in preference to GGG-A-00147.

This description establishes requirements for a 1/2-inch male drive, screwdriver blade and driving knob hand tool used for adjusting overhead valve tappets of internal combustion engines while the engine is running at idle speed.

Salient characteristics.

Material. The material shall be as specified herein.

Design. The adjusting tool shall be similar to Figure 1 and shall consist of a handle, 1/2-inch square male drive, screwdriver blade and driving knob.

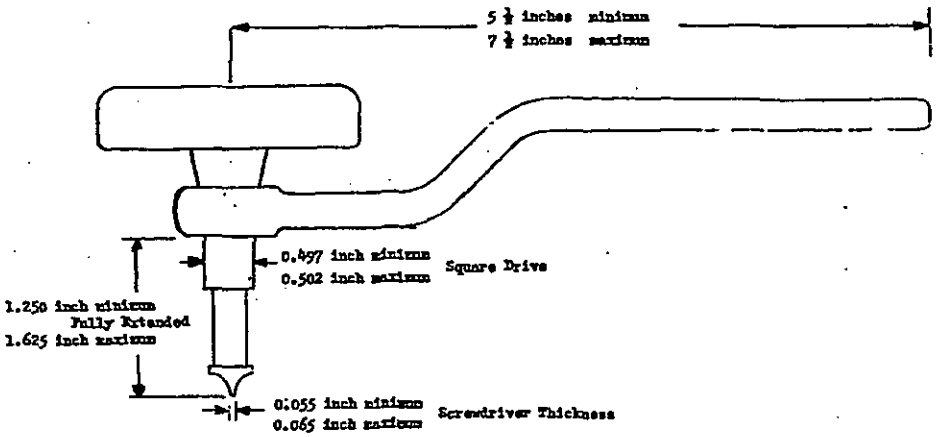


FIGURE 1. Valve Tappet Adjusting Tool

Handle. The handle shall be either offset or straight with strength and durability capable of withstanding a 300 inch-pound torque load.

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Male square drive. The square drive shall be of alloy steel, bored with a lengthwise hole to accept the screwdriver blade and shall be hardened to a Rockwell hardness of not less than 38 nor more than 54 on the "C" scale. Means shall be provided for retaining a wrench socket on the drive end, and for detaching same manually, without undue force or the use of any tool.

Screwdriver blade. The screwdriver blade shall be of steel, properly heat-treated and hardened to a Rockwell hardness of not less than 48 nor more than 58 on the "C" scale; the reading to be taken as near the tip as practicable. Means shall be provided for retaining the screwdriver within the square drive without impairing its use.

Accessories. Socket wrenches shall be furnished with 1/2 inch square drive in the following socket sizes: 1/2 inch, 9/16 inch and 5/8 inch.

Socket, socket wrenches. Socket wrenches shall be in accordance with GGG-W-641, type II, Class 2, Style A.

Identification marking. Each adjusting tool shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number and country of origin.

Finish. All surfaces shall be clean and free of pits, nodules, burrs and cracks or other detrimental defects.

Coating. The tools shall be coated to prevent rusting in accordance with accepted industry practice.

Workmanship. The adjusting tools shall be free from any defects that would affect appearance, serviceability, or durability.

QUALITY ASSURANCE PROVISIONS

Sampling for inspection and testing. Sampling shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes. The lot size shall consist of all adjusting tools offered for acceptance at one time. The sample unit is one adjusting tool.

Examination of end item for visual and dimensional requirements. Each adjusting tool shall be examined for workmanship, marking, and dimensional requirements. The inspection level shall be Level I with an acceptable quality level (AQL's) of 4.0 defects per hundred units.

Testing for hardness. The inspection level shall be S-2 with an AQL of 2.5 defects per hundred units. Testing shall be performed in accordance with ASTM E-18.

Torsion test. A torsion test shall be conducted on samples selected randomly in accordance with MIL-STD-105. Each adjusting tool shall be tested by inserting the male drive into a fixed female socket and applying a 300 inch-pound test load to the handle in accordance with Figure 2. If the tool breaks, cracks or becomes deformed, it shall be considered defective.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing supplier's employees to undue hazards from the recovered materials.

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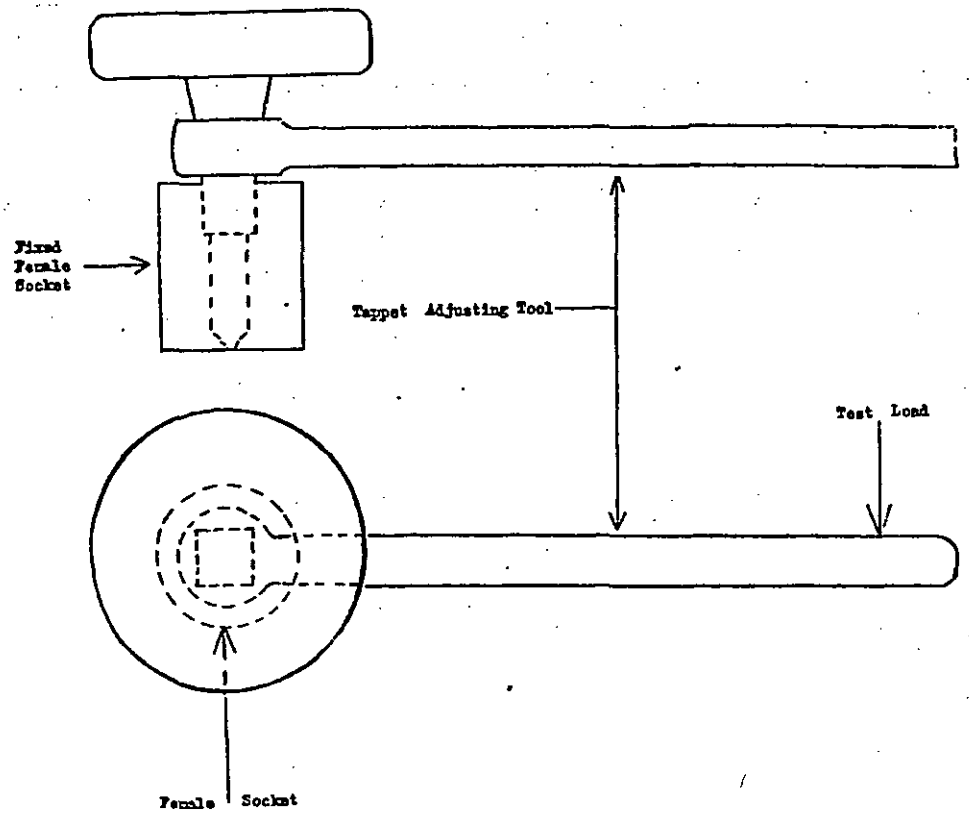


Figure 2. Application of load for torsion testing.

Preservation, packaging, packing, labeling and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes. Purchasers should specify if socket wrenches are required. The issue of referenced documents in effect on the date of invitation for bid or request for proposals shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the Contracting Officer.

ASTM test method is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

PREPARING ACTIVITY

GSA-FSS

NSN'S 5120-00-293-0595 - with accessories

Socket, socket wrench:

NSN's 5120-00-237-0984 - 1/2 inch
5120-00-189-7932 - 9/16 inch
5120-00-189-7946 - 5/8 inch

NSN 5120-00-293-0662 - without accessories

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